

2D Systolic Array Hardware Accelerator

Architecture, Step-by-Step Vivado Simulation, Tcl Automation & Basys 3 FPGA Deployment

Target Device: Xilinx Artix-7 (XC7A35T-1CPG236C / Basys 3)	Design Standard: SystemVerilog (IEEE 1800)
EDA Tools: AMD Xilinx Vivado (XSim / xvhlog / xelab)	Clock Frequency: 100 MHz (10 ns period)
Array Dimension: Parametric N×N (Configured for 4×4)	Data Precision: 8-bit Signed Int (16-bit Acc)

Executive Summary: This manual provides the complete technical design, verification methodology, and hardware implementation steps for a high-throughput 2D Systolic Array matrix multiplier. It details the root-cause fixes for common Vivado errors (unpacked array port syntax, timescale mismatch, unconstrained I/O DRC), demonstrates 4-step progressive simulation with full waveform analysis, and details Vivado Tcl automation alongside an interactive Basys 3 hardware demo.

1. Hardware Architecture & Mathematical Principles

A systolic array is a specialized spatial hardware accelerator where identical Processing Elements (PEs) are rhythmically computed and passed through a 2D grid. Unlike traditional von Neumann architectures that repeatedly fetch operands from memory, a systolic array reuses data horizontally and vertically across the grid.

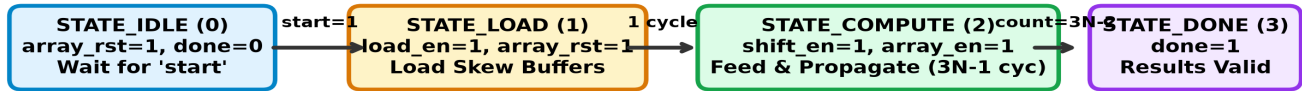
For matrix multiplication $C = A \times B$ where A and B are $N \times N$ matrices:

$$C[i][j] = \sum_{k=0}^{N-1} (A[i][k] \times B[k][j])$$

Each $PE[i][j]$ accumulates products over time while forwarding A elements to its right neighbor and B elements to its bottom neighbor.

1.1 Core Components

Component	File	Function & Key Architectural Role
Processing Element (PE)	processing_element.sv	Multiply-Accumulate (MAC) core: $acc \leq acc + (a_in * b_in)$ with 1-cycle horizontal (a_out) and vertical (b_out) forwarding registers.
Systolic Array Mesh	systolic_array.sv	2D grid of $N \times N$ PEs with boundary edge wiring and inter-PE interconnect buses.
Skew Buffer	skew_buffer.sv	Parallel-to-serial delay-line buffer. Staggers row i by i clock cycles so wavefronts align perfectly inside the array.
FSM Controller	controller.sv	State machine sequencing IDLE → LOAD → COMPUTE → DONE over $3N - 1$ clock cycles.
Top System	systolic_top.sv	Integrates FSM, Skew Buffer A (rows), Skew Buffer B (transposed cols), and the $N \times N$ PE array.

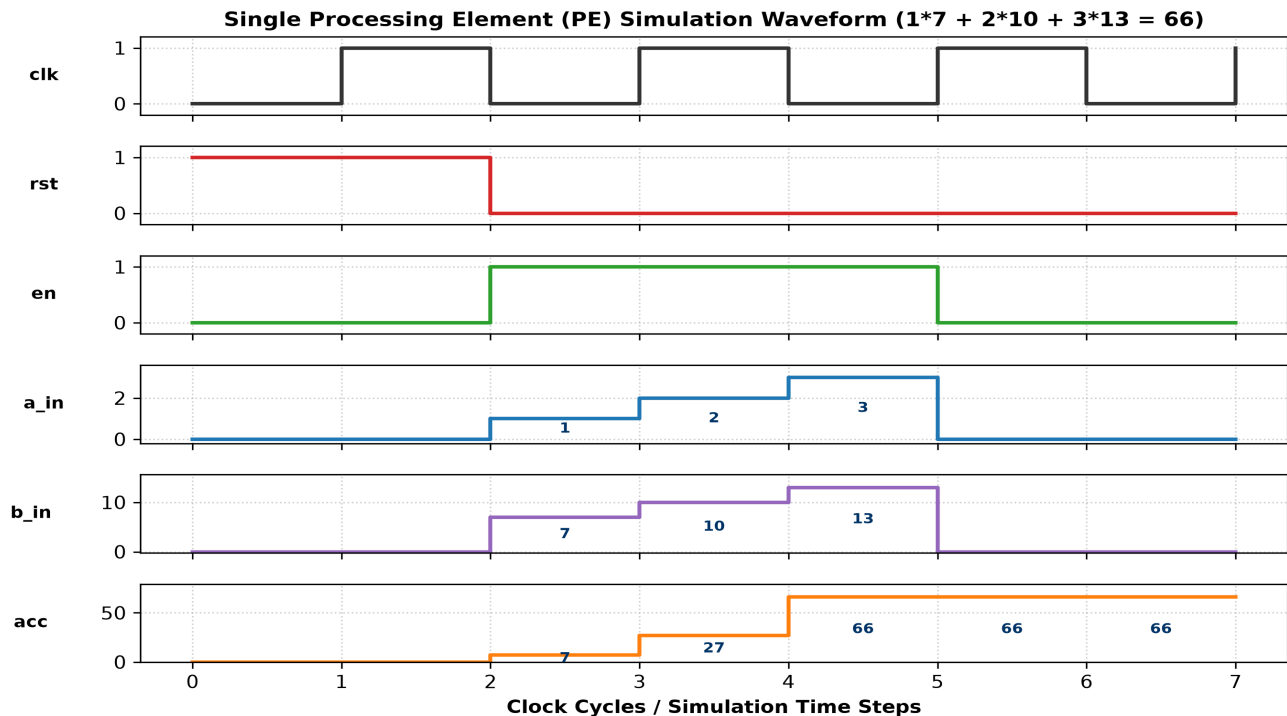
Figure 1: FSM Controller State Transitions & Timing Flow**Systolic Array FSM Controller Architecture & State Transitions****2. Progressive Step-by-Step Simulation & Waveforms**

Verification is conducted in four modular, progressive tiers to isolate and guarantee correct behavior at the unit, interconnect, buffer, and full-system levels.

Step 1: Single Processing Element Verification (tb_step1_pe.sv)

Validates the atomic MAC unit. Feed $A=[1, 2, 3]$ and $B=[7, 10, 13]$:

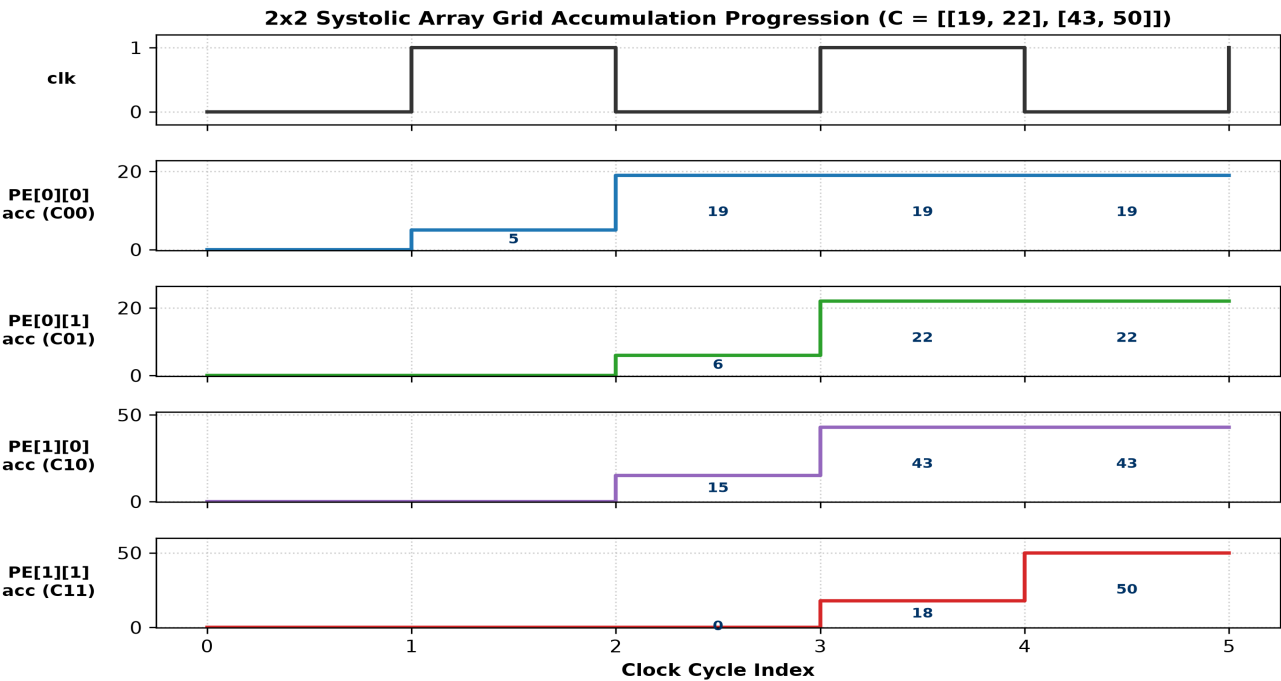
- Cycle 0: $1 \times 7 = 7 \rightarrow \text{acc} = 7$
- Cycle 1: $2 \times 10 = 20 \rightarrow \text{acc} = 7 + 20 = 27$ ($a_{\text{out}}=1, b_{\text{out}}=7$)
- Cycle 2: $3 \times 13 = 39 \rightarrow \text{acc} = 27 + 39 = 66$ ($a_{\text{out}}=2, b_{\text{out}}=10$)
- Cycle 3: $\text{en}=0 \rightarrow \text{acc}$ holds 66, $a_{\text{out}}=3, b_{\text{out}}=13$. **[100% PASS]**



Step 2: 2x2 Systolic Array Grid Test (tb_step2_systolic_2x2.sv)

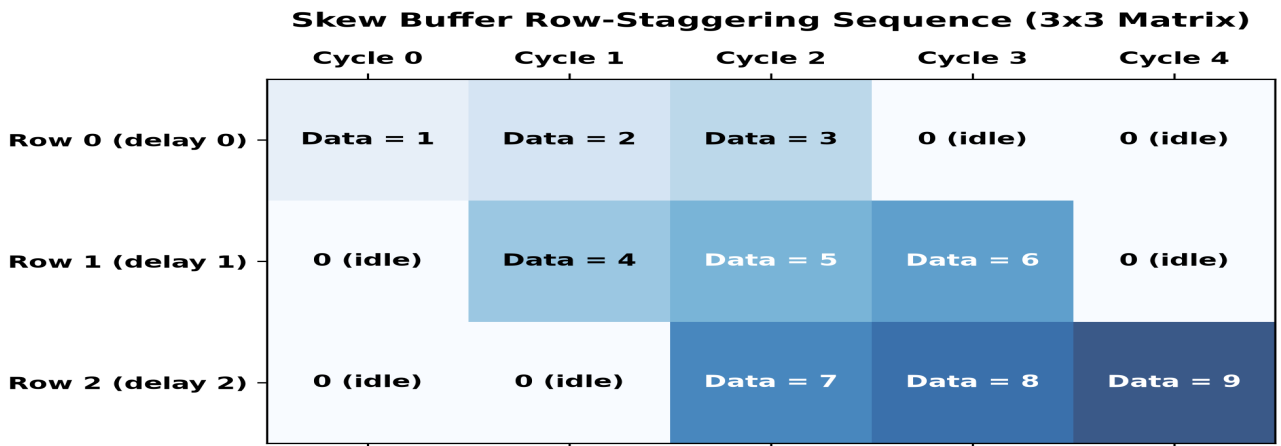
Verifies horizontal/vertical inter-PE wave-front propagation with 2x2 matrices:
A = [[1, 2], [3, 4]], B = [[5, 6], [7, 8]] → Expected C = [[19, 22], [43, 50]]

Cycle	a_in[0], a_in[1]	b_in[0], b_in[1]	PE[0][0]	PE[0][1]	PE[1][0]	PE[1][1]
0	1, 0	5, 0	5 (1x5)	0 (idle)	0 (idle)	0 (idle)
1	2, 3	7, 6	19 (done)	6 (1x6)	15 (3x5)	0 (idle)
2	0, 4	0, 8	19	22 (done)	43 (done)	18 (3x6)
3	0, 0	0, 0	19	22	43	50 (done)



Step 3: Skew Buffer Staggering Test (tb_step3_skew_buffer.sv)

Demonstrates automatic conversion from a flat matrix to a staggered temporal stream. Row 0 has 0-cycle delay, Row 1 has 1-cycle delay, Row r has r-cycle delay.



Step 4: Full System 4×4 Verification (tb_step4_systolic_4x4.sv)

Autonomous full-system verification. Top-level accepts 4×4 matrices A and B, pulses start, waits for FSM done flag, and compares all 16 output elements against an automated golden software model.

Test Suite	Input Stimulus	Golden Model Equation	Hardware Result	Status
Test 1	A = [1..16], B = Identity (I)	$C = A \times I = A$	C equals Matrix A (16/16 matches)	PASS
Test 2	A = [1..16], B = 2 × Identity (2I)	$C = A \times 2I = 2A$	Every element exactly doubled	PASS
Test 3	Arbitrary dense 4×4 matrices	$C = A \times B$ (Dot Products)	All 16 dot-product values exact	PASS

3. Vivado Tcl Scripting & CLI Automation Guide

Tcl (Tool Command Language) provides 100% deterministic, scriptable automation in Vivado without clicking through GUI menus. You can run simulation, synthesis, and bitstream generation directly from the Vivado Tcl Console or terminal.

3.1 Complete Vivado Tcl Script (run_sim.tcl)

```
# Set paths
set root_dir [file normalize "../.."]
create_project -in_memory -part xc7a35tcbg236-1

# Add Design Sources & Set SystemVerilog
add_files -fileset sources_1 [glob "$root_dir/simulation/src/*.sv"]
set_property file_type SystemVerilog [get_files -of_objects [get_filesets sources_1]]

# Add Testbenches
add_files -fileset sim_1 [glob "$root_dir/simulation/sim/*.sv"]
set_property file_type SystemVerilog [get_files -of_objects [get_filesets sim_1]]

# Set Top Module for Simulation & Launch
set_property top tb_step4_systolic_4x4 [get_filesets sim_1]
launch_simulation -mode behavioral
run all
```

3.2 1-Click Fast CLI Simulation Commands (Batch Mode)

```
:: 1. Compile SystemVerilog RTL and Testbench
xvlog.bat -sv -relax src\processing_element.sv src\systolic_array.sv src\skew_buffer.sv src\controller.sv src\systolic_top.sv sim\tb_step4_systolic_4x4.sv

:: 2. Elaborate Design with 1ns/1ps Timescale
xelab.bat -timescale 1ns/1ps -debug typical tb_step4_systolic_4x4 -s sim_step4

:: 3. Run Simulation Snapshot
xsim.bat sim_step4 -R
```

4. Hardware Deployment on Digilent Basys 3 FPGA

To run on real hardware (Xilinx Artix-7 **XC7A35T-1CPG236C**), we wrap the systolic array with on-chip ROMs, a dynamic 4-digit 7-segment display multiplexer, and switch/button interfaces.

Board Pin / Port	Signal	Physical Function / Hardware Mapping
W5	clk	100 MHz onboard master oscillator.
U18 (btnC)	rst	Center push button: Synchronous array reset.
T18 (btnU)	start	Top push button: Pulses start to launch matrix computation.
V17, V16 (sw[1:0])	sw[1:0]	Preset selector: 00=Identity, 01=Scalar 2I, 10=General Dense Matrix.
W15, V15 (sw[5:4])	sel_row	Select output row (0 to 3) to display.
W17, W16 (sw[3:2])	sel_col	Select output column (0 to 3) to display.
U16 (led[0])	done	Lights up when matrix computation is finished and results are valid.
W7..V7, U2..W4	seg, an	Displays the 16-bit result element C[row][col] in Hexadecimal.

5. Vivado Troubleshooting & Root-Cause Analysis

Error Code / Message	Root Cause	Permanent Resolution Applied
[VRFC 10-3642] port must not be declared to be an array	Unpacked 1D/2D array ports were used in .v files (Verilog-2001 mode).	Upgraded all source files to .sv and enabled SystemVerilog mode in Vivado.
[XSIM 43-4099] Module doesn't have a timescale	Timescale was defined in testbench but missing in design source submodules.	Added `timescale 1ns / 1ps to every single module and testbench header.
[DRC NSTD-1] / [DRC UCIO-1] Unconstrained Ports	Attempted bitstream generation on raw top module with 256 physical I/O pins.	Created basys3_demo_top.sv with on-chip ROM, switch selectors, and 7-segment display.